

Task 01 — Python Programming, Data Analysis & Visualization

Video Game Sales Analysis

Program: AI/ML Internship Program

Task: Task 01

Student Name: [Enter your name]

Domain: AI/ML

Submission Date: 8 September 2026

Note: The GitHub repository URL and student name were not available in the supplied project materials, so placeholders are retained rather than inventing them.

1. Introduction

This project analyzes historical video game sales data to understand trends in the gaming market, compare regional sales, and examine performance across platforms, genres, and publishers.

2. Objective

The objective is to complete a practical data-analysis workflow using Python: load and inspect the dataset, clean the data, perform exploratory and statistical analysis, create meaningful visualizations, generate insights, and communicate conclusions.

3. Dataset Name

Video Game Sales dataset, loaded from the file `vgsales.csv`.

4. Dataset Source

Kaggle dataset used in the analysis notebook. The separate CSV file was not included in the uploaded session files, so no replacement dataset has been fabricated.

5. Dataset Description

The dataset contains video game records with game name, platform, release year, genre, publisher, regional sales, and global sales information.

6. Dataset Size

The dataset contains 16,598 rows and 11 columns.

7. Features / Columns Description

Rank: ranking value; Name: game title; Platform: gaming platform; Year: release year; Genre: game genre; Publisher: publisher; NA_Sales: North American sales; EU_Sales: European sales; JP_Sales: Japanese sales; Other_Sales: sales from other regions; Global_Sales: worldwide sales. Numerical variables are used for statistical analysis, while categorical variables support grouping and comparisons. There is no full date-time column.

8. Methodology

The workflow follows Dataset Selection → Loading → Understanding → Cleaning → EDA → Statistics → Visualization → Insights → Conclusion → Documentation.

9. Data Loading

The notebook imports Pandas, NumPy, Matplotlib, and Seaborn and loads `vgsales.csv` with Pandas. It displays the first five rows, last five rows, and a random sample.

10. Data Cleaning

Missing Publisher values were replaced with `Unknown` to preserve records without inventing a publisher. Missing Year values were not filled with an unsupported year; instead, rows without a year were excluded from the separate time-analysis dataframe. Duplicate rows were checked and none were found. Sales columns were checked for negative values and none were found.

11. Exploratory Data Analysis

The analysis examines dataset structure, missing values, year coverage, game releases over time, global sales over time, average sales, platform performance, genre performance, publisher performance, and regional sales.

12. Statistical Analysis

Descriptive statistics were calculated for the sales variables, including mean, median, minimum, maximum, mode, range, standard deviation, and variance. Correlations were also calculated. NA vs Global is approximately 0.94, EU vs Global approximately 0.90, NA vs EU approximately 0.77, and JP vs Global approximately 0.61.

13. Data Visualizations

The notebook contains the required visualization types: histogram, bar chart, pie chart, scatter plot, line charts, box plot, correlation heatmap, count plot, pair plot, and violin plot. Visualizations use meaningful variables and include titles, labels where applicable, and appropriate figure sizes.

14. Code Screenshots

Selected code extracts from the actual notebook are included below.

15. Output Screenshots

The report uses outputs already stored in the executed notebook, including descriptive statistics, correlations, and top-game results.

16. Graph Screenshots

Selected graphs generated by the notebook are included below.

17. Major Findings

Video game sales are highly uneven. Wii Sports is the highest-selling title at 82.74 million global sales. North America and Europe show especially strong relationships with global sales. The number of releases rises substantially over time and is highest around 2008–2009.

18. At Least 10 Insights

1) Wii Sports is the highest-selling game globally at 82.74 million. 2) Nintendo appears repeatedly among the top-selling games. 3) Game releases increase substantially over time and peak around 2008–2009. 4) Median global sales (0.17m) are much lower than mean global sales (0.54m), indicating right-skew. 5) North America has the highest average regional sales. 6) NA sales strongly correlate with global sales (~0.94). 7) EU sales strongly correlate with global sales (~0.90). 8) JP sales have a positive but weaker correlation with global sales (~0.61). 9) No duplicate rows are present. 10) No negative sales values are present.

19. Conclusion

The analysis shows that video game sales are highly uneven: most games have relatively low sales, while a small number of titles achieve very high global sales. Nintendo titles appear strongly among the best-selling games, and North American and European sales show especially strong relationships with global sales. Game releases increased considerably over time, with the highest concentration around 2008–2009. Overall, the dataset provides useful evidence about differences in game performance across years, regions, genres, platforms, and publishers.

20. GitHub Repository Link

[Paste your GitHub repository link here after creating the repository. The notebook, dataset, README, visualization images, and PDF report should be included in the repository.]

Code Screenshots

```
code_01_loading

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("vgsales.csv")
```

Code 01 Loading

```
code_02_cleaning

df["Publisher"] = df["Publisher"].fillna("Unknown")
df_year = df.dropna(subset=["year"]).copy()
df_year["year"] = df_year["year"].astype(int)
df.duplicated().sum()
```

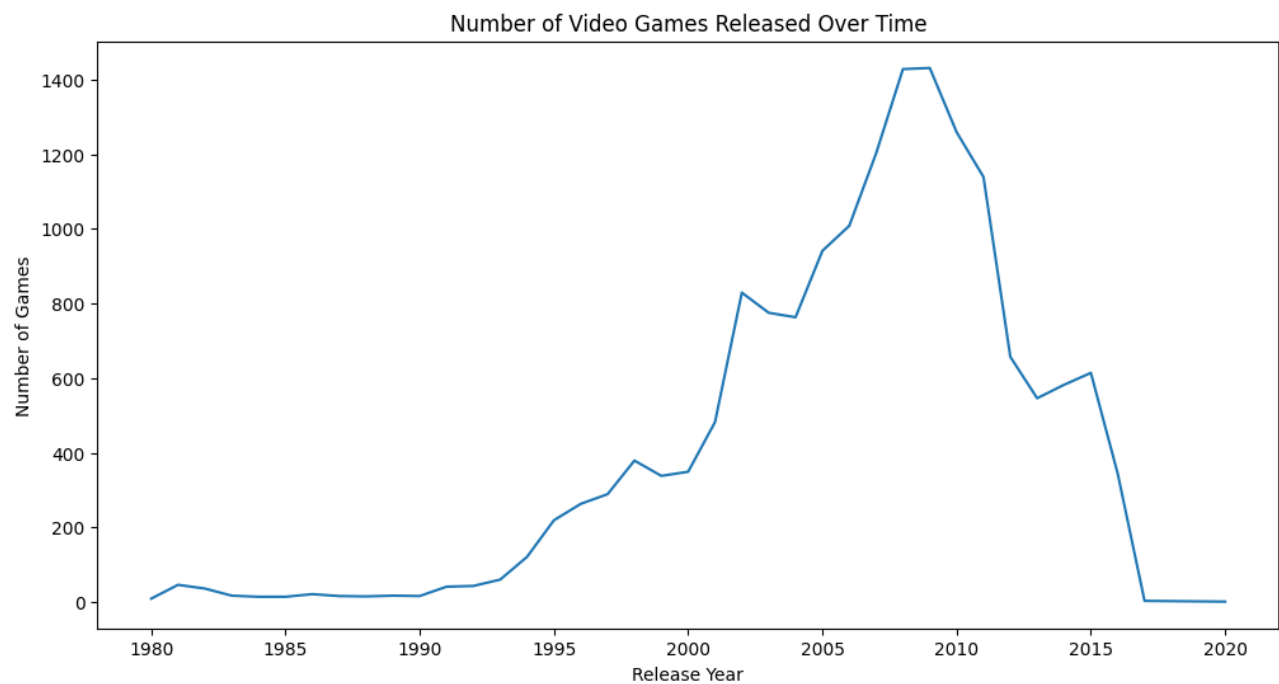
Code 02 Cleaning

```
code_03_visualization

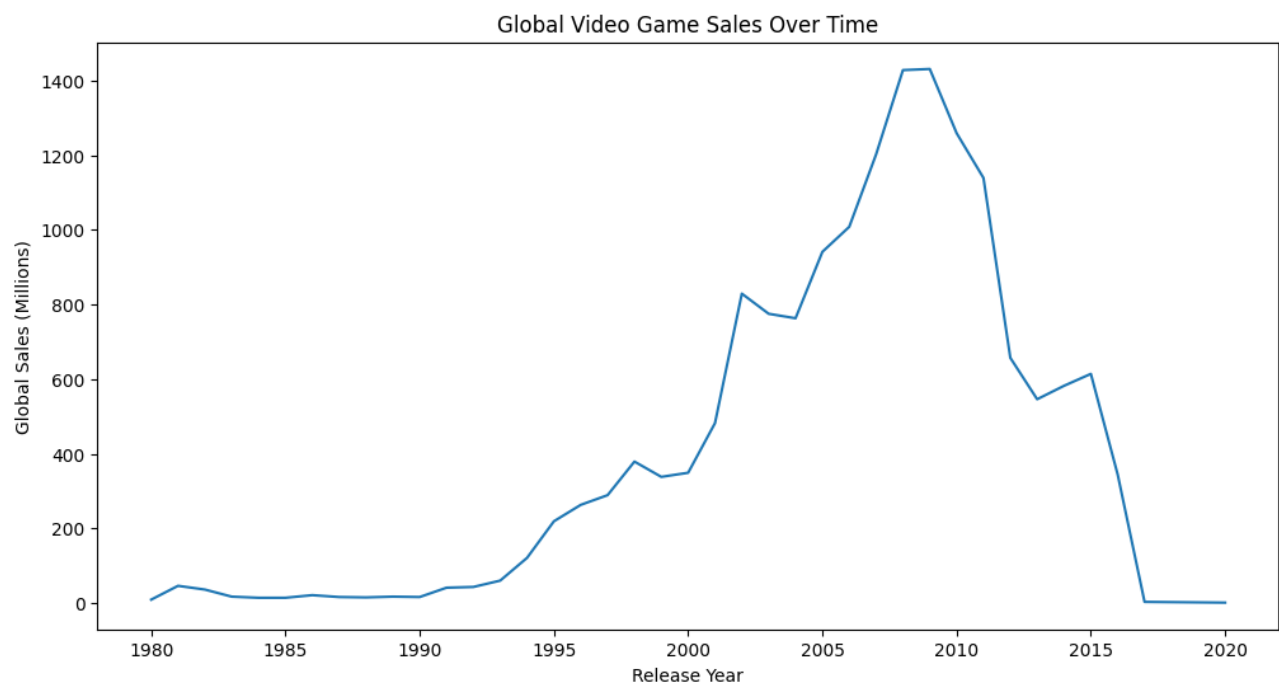
sns.lineplot(
sns.barplot(
sns.histplot(
sns.boxplot(
sns.scatterplot(
sns.heatmap(
sns.countplot(
sns.violinplot(
sns.pairplot(
```

Code 03 Visualization

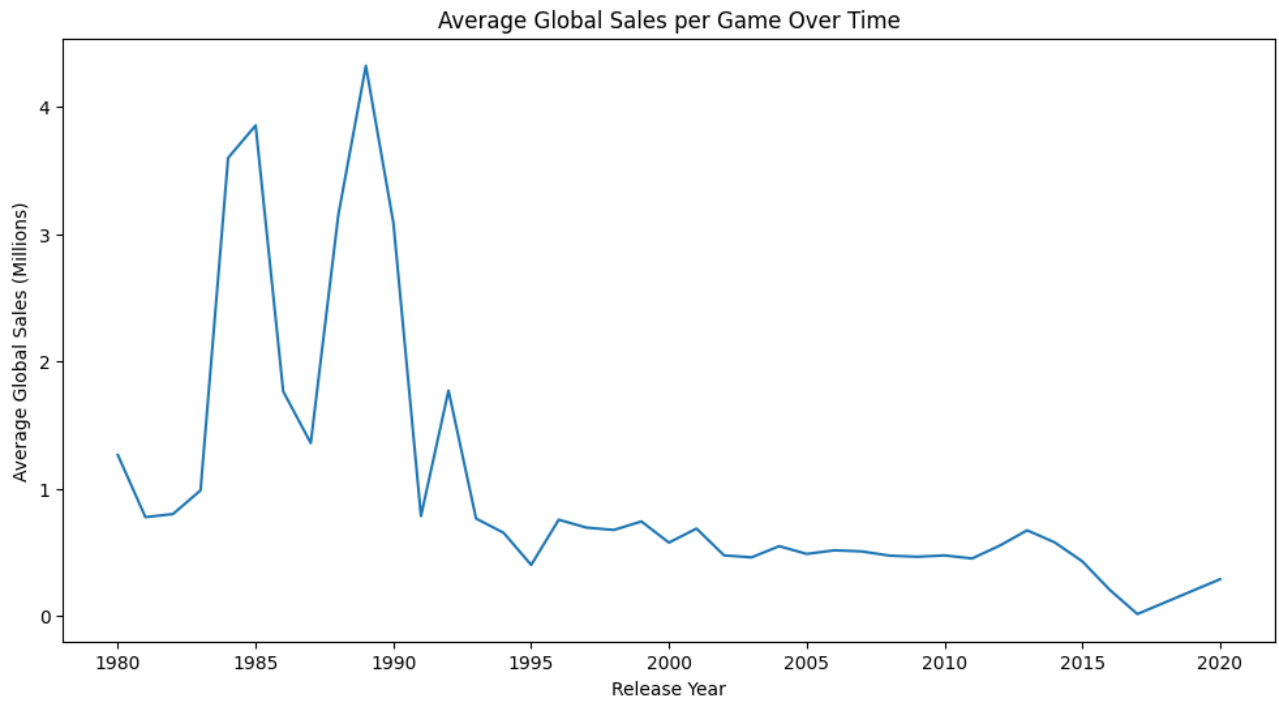
Graph Screenshots



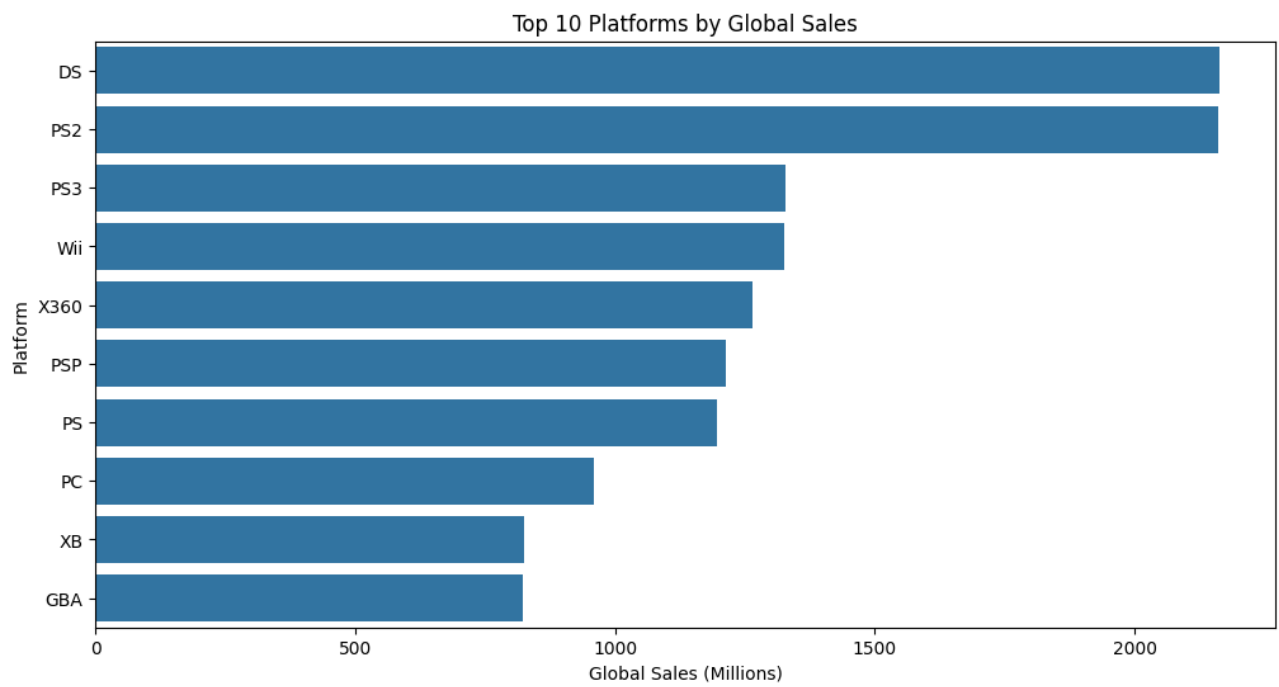
Visualization 1



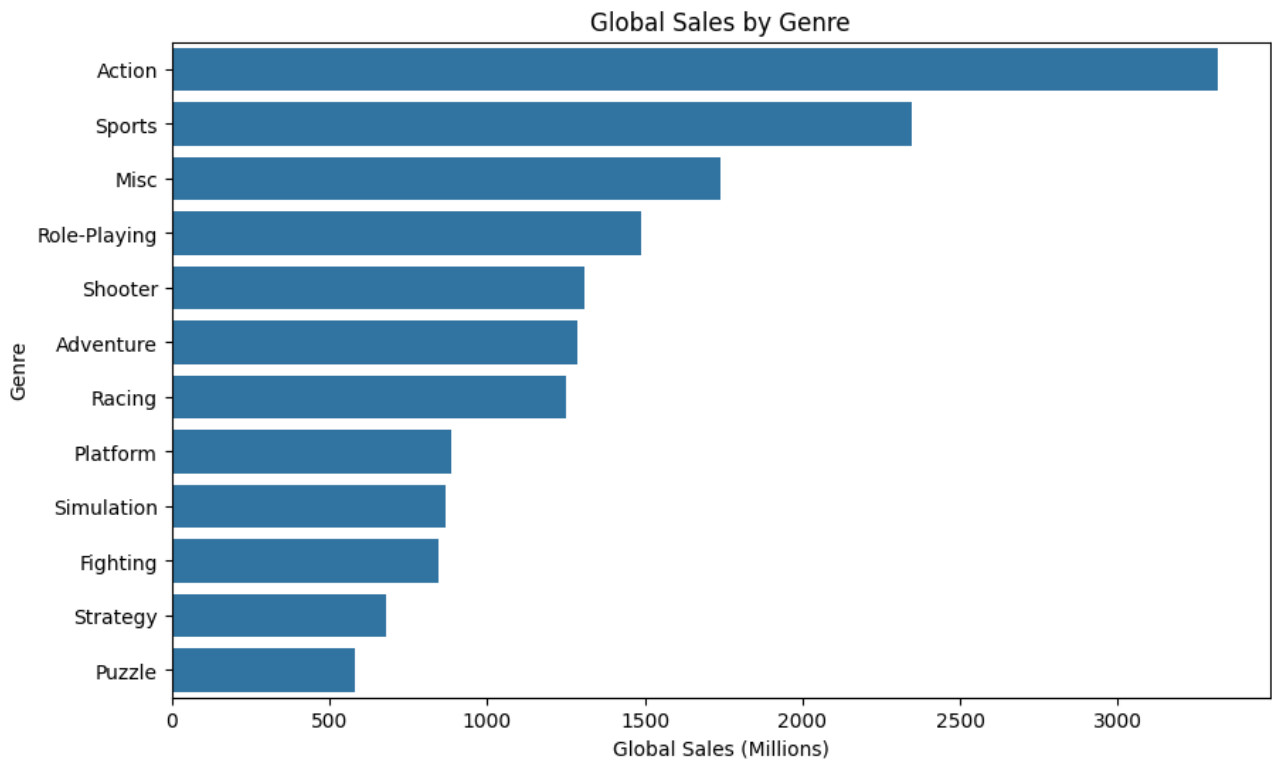
Visualization 2



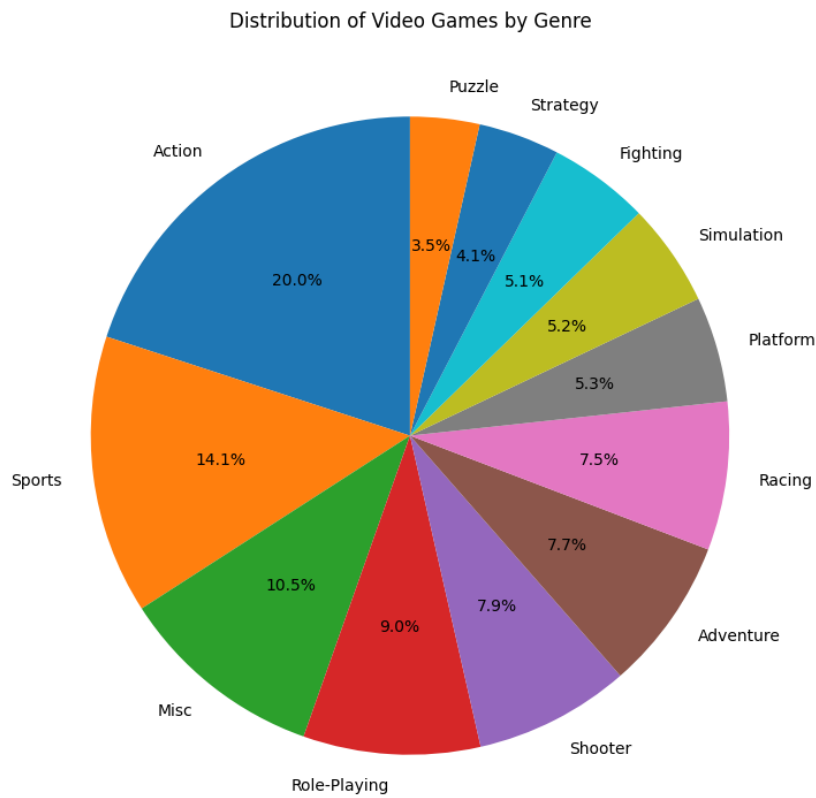
Visualization 3



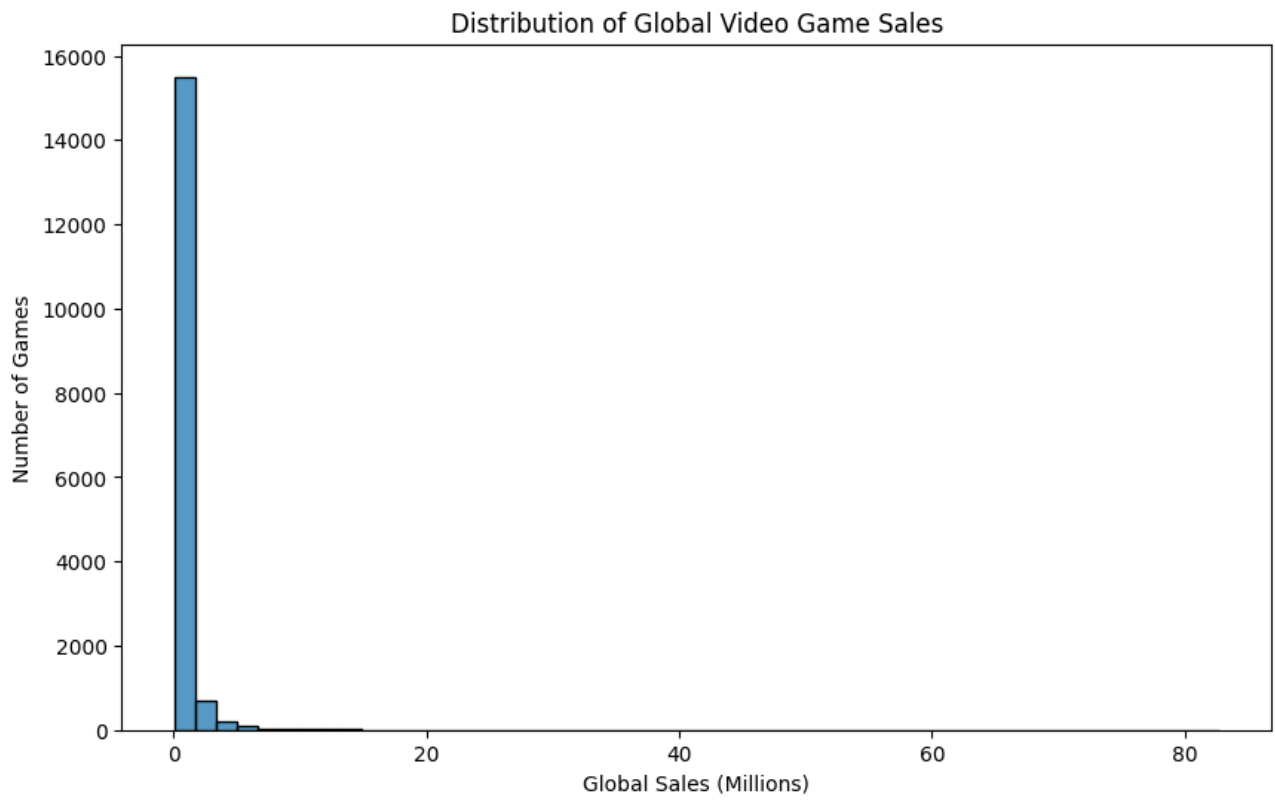
Visualization 4



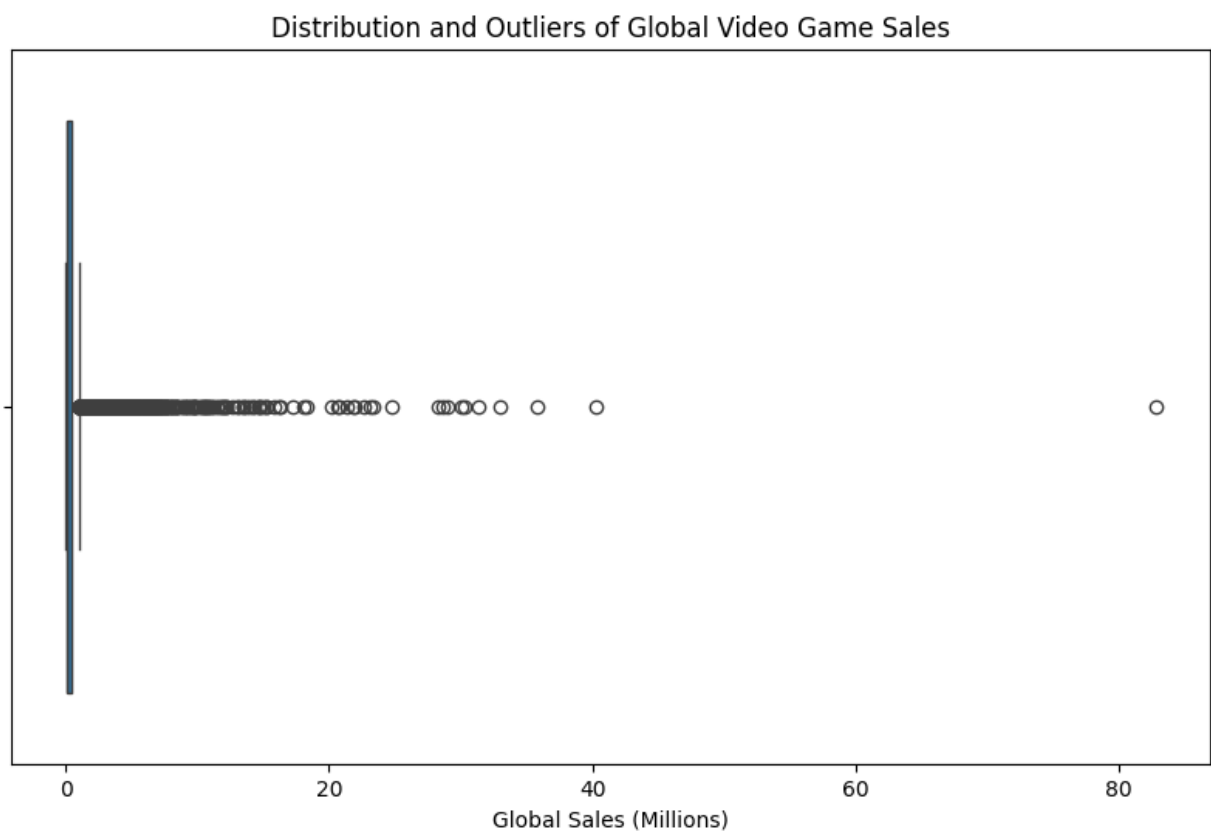
Visualization 5



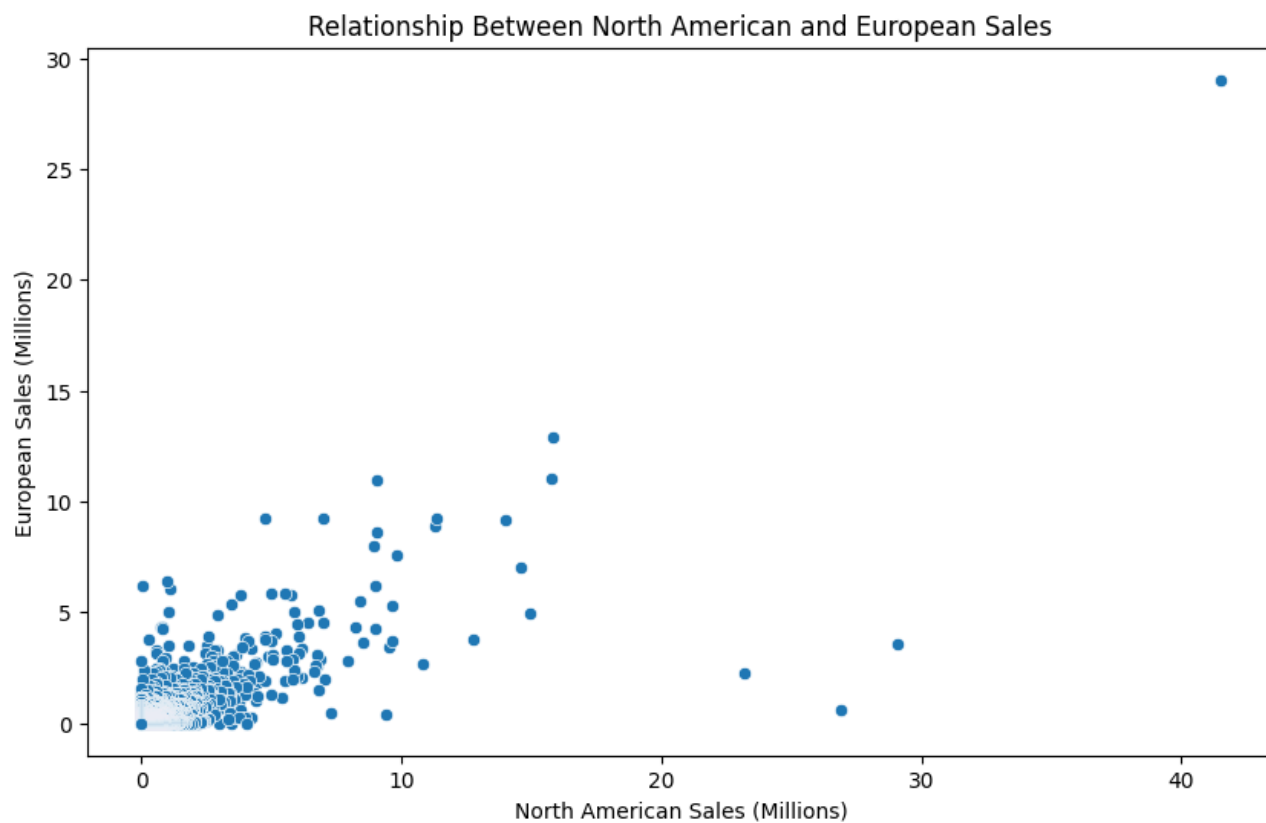
Visualization 6



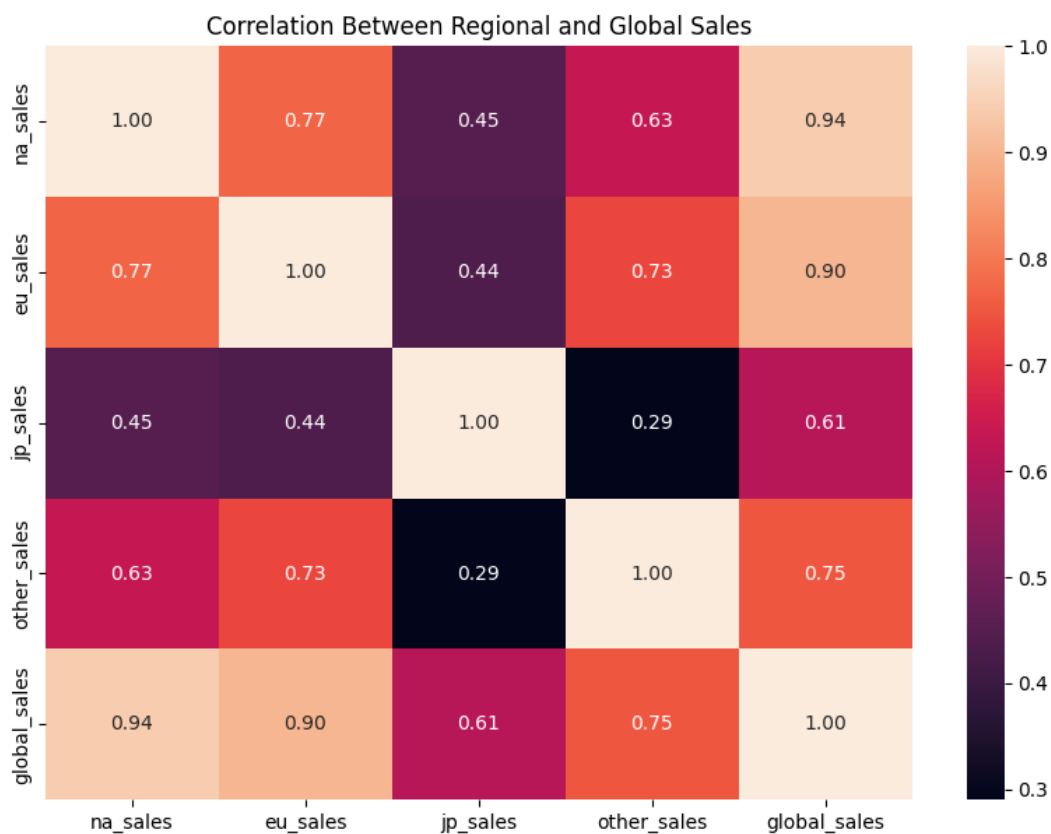
Visualization 7



Visualization 8



Visualization 9



Visualization 10